

Amaan Mansuri

📍 New York, NY 📞 +1 (562) 209-6787 ✉️ amaan.mansuri@nyu.edu **in** [amaanmansuri](#)  [Amaan165](#)

EDUCATION

New York University – Center for Data Science

Master of Science in Data Science

New York, USA

Expected May 2026

Nirma University – Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Ahmedabad, India

Sept 2020 - Jun 2024

EXPERIENCE

AI/ML Research Engineer

United Nations Development Programme (UNDP)

New York, USA

Sep 2025 – Present

- Developing and deploying computer vision models to detect target marine species in real time from underwater video, enabling researchers to monitor ecosystems while allowing tourists to explore via VR without disturbing habitats.
- Building the end-to-end pipeline: data curation and labeling, class schema with domain experts, augmentation and color-correction for turbidity/low-light; model training with **YOLOv11** and multi-object tracking.

Research Assistant

Hartley Lab, Department of Psychology, New York University

New York, USA

Mar 2025 – Present

- Developed a web scraping pipeline to collect and process over **15,000** metadata entries from books and media sources, enhancing stimuli selection efficiency by **60%**.
- Proposed and implemented **knowledge graph-based** models to simulate semantic relationships across age groups, boosting prediction accuracy in reasoning tasks by **25%**.
- Created visualizations and result summaries for IRB reports and manuscript drafts, improving clarity and comprehension for non-technical reviewers by **30%** (based on internal lab evaluation).

Software Development Engineer Intern

Johnson Controls-Hitachi

Kadi, India

Jan 2024 - Jun 2024

- Engineered a Bluetooth-based communication system for Hitachi ACs using UART, boosting NLOS stability by **33%** and expanding range from **7m to 55m**.
- Built an Android app replicating IR remote controls, improving device response time by **40%** through optimized signal handling.

PROJECTS

flight-wx: Flight Delay & Weather Monitoring System (Big Data + API Integration)

[Github](#) 

- Developed a real-time flight delay tracking system integrating FAA and NOAA APIs, supporting flexible city-to-airport queries and resolving metadata for **400+ U.S. airports** with high accuracy and reliability.
- Engineered modular CLI tools with batch-mode ingestion, intelligent caching, and multi-airport resolution logic, reducing redundant API hits by **70%** and improving end-to-end ingestion speed by **3x**.

Multi-Horizon Stock Price Prediction (LSTM + FinBERT)

[Github](#)  [Report](#) 

- Built a deep learning pipeline using **LSTM networks** to forecast 1-day and 3-day prices for **26 major stocks**, integrating **FinBERT-based sentiment scores** to boost accuracy by up to **18.57%** and directional accuracy by **12%**.
- Evaluated four feature sets (price, technicals, sentiment, combined) using **RMSE**, **MAE**, **MAPE**, and **R²**; achieved best-case **R² = 0.969** (AAPL, 1-day); optimized training with **ReduceLRonPlateau**, **EarlyStopping**, and **TensorBoard**.

Simulated CCAR Stress Test (Machine Learning + API Integration)

[Github](#) 

- Achieved a CCAR-style stress testing pipeline to simulate **macroeconomic shock impact** (unemployment, inflation, GDP decline) on **retail loan defaults**, modeling forward-looking risk exposure.
- Trained a borrower-level **Probability of Default (PD)** model and computed **Expected Loss = PD × EAD × LGD**, estimating capital reserve needs under adverse conditions.

Humor Detection (Machine Learning + NLP)

[Github](#) 

- Designed an NLP-based humor classification pipeline using **BoW**, **TF-IDF**, and **Word2Vec**, trained on a labeled corpus of **200k texts**, achieving **88%** accuracy and cutting misclassification by **20%**.
- Extended the pipeline by fine-tuning an LLM on the **Humicroedit** dataset to generate humorous versions of non-humorous text; incorporated automatic un-funny word detection and transformation using generative decoding.

SKILLS

Languages: Python, SQL, Java, C++, C, Bash, HTML, CSS, TypeScript, Solidity.

Frameworks & Libraries: NumPy, Pandas, scikit-learn, Matplotlib, XGBoost, TensorFlow, Keras, PyTorch, Transformers, NLTK, spaCy, LangChain, Hugging Face, Django, Flask, Hadoop, Spark, Dask, Git, JAX, OpenCV.

Tools & Platforms: Tableau, Power BI, Docker, TensorBoard, Airflow, AWS, GCP, Unix/Linux, Anaconda, FastAPI, Selenium.

COURSEWORK

Machine Learning, Deep Learning, Big Data, Programming for Scientific Computing, Data Mining, Linear Algebra, Probability and Statistics, Database Management Systems, Computer Architecture, Operating Systems.